Problem statement : Ride & Handling model for vehicle config 6x2 → front axle, (f_1) & rear axle (r_1, r_2) → NRS non reactive suspension tandem rear → output at the seat level. capture Ride → rms acceleration, Handling vehicle roll, tail sway parameter, tyre slip angle.

NRS → load sharing, it doesnot cause any roll, pitch due to moments by powertrain.

$$R_1 \xrightarrow{4\text{ bar}} \underset{BCL}{\rightarrow} \xrightarrow{4\text{ bar}} R_2$$

4 bar equations.

$$\beta_2 \rightarrow \psi \rightarrow \beta_3$$

$$\theta_4 = 2\tan^{-1}\left(\frac{L_2 S\theta_2 - L_3 S\gamma}{L_2 C\theta_2 + L_4 - L_1 - L_3 \cos\gamma}\right)$$

$$BD = \sqrt{L_1^2 + L_2^2 - 2L_1L_2 C\theta_2}$$

$$\gamma = \cos^{-1}\left[\frac{L_3^2 + L_4^2 - BD^2}{2L_3L_4}\right]$$

$$\theta_3 = 2\tan^{-1}\left(\frac{-L_2 S\theta_2 + L_4 S\gamma}{L_1 + L_3 - L_2 S\theta_2 - L_4 C\gamma}\right)$$

NRS kinematics.

$$\beta_2(t) \xrightarrow{4\text{bar}\#1} \psi(t) \xrightarrow{\because \text{ //gm}} \psi(t) \xrightarrow{4\text{bar}\#3} \beta_3(t).$$

Vertical displacement from NRS geometry

$z_0 = L_D \sin\alpha$ initial vertical coord of spring centre

$z_1 = L_D \sin(\alpha+\beta)$ new vertical coord after rotation.

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Date :

1. Seat vertical motion - DOF1.

$$M_{ch}\ddot{z}_{ch} + k_{ch}(z_{ch} - (z_c - e\theta)) + c_{ch}(\dot{z}_{ch} - (\dot{z}_c - e\dot{\theta})) = 0.$$

2. Cab vertical motion - DOF2

$$M_{dr}\ddot{z}_{dr} + k_{dr}(z_{dr} - (z_c - e\theta)) + c_{dr}(\dot{z}_{dr} - (\dot{z}_c - e\dot{\theta})) = 0$$

3. Chassis bounce equation → DOF-3 $\quad 2 \to r_1 (m_1)$, $3 \to r_2$

$$M_2\ddot{z}_c + k_f(z_c - a\theta - z_{1f}) + c_f(\dot{z}_c - a\dot{\theta} - \dot{z}_{1f})$$
$$+ k_2(z_c - z_2 - \beta_{L2}L_{DL2} - \beta_{R2}L_{DR2} + l_2\theta)$$
$$+ c_2(\dot{z}_c - \dot{z}_2 + l_2\dot{\theta})$$
$$+ k_3(z_c - z_3 - \beta_{L3}L_{DL3} - \beta_{R3}L_{DR3} + l_3\theta)$$
$$+ c_3(\dot{z}_c - \dot{z}_3 + l_3\dot{\theta}) = 0.$$

4. Chassis pitch equation → DOF 4.

$$I_{yy}\ddot{\theta} - a[k_f(z_c - a\theta - z_{1f}) + c_f(\dot{z}_c - a\dot{\theta} - \dot{z}_{1f})]$$
$$+ l_2[k_2(z_c - z_2 - \beta_{L2}L_{DL2} - \beta_{R2}L_{DL2} + l_2\theta)$$
$$+ c_2(\dot{z}_c - \dot{z}_2 + l_2\dot{\theta})]$$
$$+ l_3[k_3(z_c - z_3 - \beta_{L3}L_{DL3} - \beta_{R3}L_{DL3} + l_3\theta)$$
$$+ c_3(\dot{z}_c - \dot{z}_3 + l_3\dot{\theta})]$$

5. Front axle → unsprung mass – DOF-5

$$M_{1f}\ddot{z}_{1f} + k_f(z_{1f} - (z_c - a\theta))$$
$$+ c_f(\dot{z}_{1f} - (\dot{z}_c - a\dot{\theta}))$$
$$+ k_{tf}(z_{1f} - q_f) = 0$$

6. Rear – 1 → DOF-6

$$M_2\ddot{z}_2 - k_2(z_c - z_2 - \beta_{L2} L_{DL2} - \beta_{R2} L_{DR2} + l_2\theta)$$
$$- c_2(\dot{z}_c - \dot{z}_2 + l_2\dot{\theta})$$
$$+ k_{t2}\left(z_2 - \left(\frac{z_{rL2} + z_{rR2}}{2}\right)\right) = 0$$

7. Rear – 2 → DOF-7

$$M_3\ddot{z}_3 - k_3(z_c - z_3 - \beta_{L3} - L_{DL3} - \beta_{R3} L_{DR3}) + l_3\theta$$
$$- c_3(\dot{z}_c - \dot{z}_3 + l_3\dot{\theta})$$
$$+ k_{t2}\left(z_3 - \left(\frac{z_{rL3} + z_{rR3}}{2}\right)\right) = 0$$

So, β_{L2} β_{L3} L_{DL2} L_{DR2}

β_{R2} β_{R3} L_{DR3} L_{DR3}

↓

4 eqn are solved to get these

Shock absorber :

$$F_s = -c\dot{r} + k(r-L) + F_{preload} \qquad r \neq L$$

$$F_s = \text{preload}.$$

Roll dynamics. → DOF 8

$$I_{xx}\ddot{\phi} + \sum_{i \in f,m,r} (k_{\phi i}\phi + c_{\phi,i}\dot{\phi}) + \sum_{i \in f,m,r} \underbrace{F_{yi}\, h_{rc,i}}_{\text{might be excluded (need to check if needed)}} = 0$$

$k, c \rightarrow$ elastic, viscous roll at axles.

lateral tire forces acting at roll centre heights $h_{rc,i}$

Lateral & yaw dynamics. → DOF 9 & DOF-10
↳ v_y ↳ ψ.

$$m(\dot{v}_y + ur) = F_{y,f} + F_{y,m} + F_{y,r}$$

$$I_{zz}\ddot{r} = aF_{y,f} - l_2 F_{ym} - l_3 F_{y,r}$$

$$\dot{\psi} = r$$

Tire slip angles

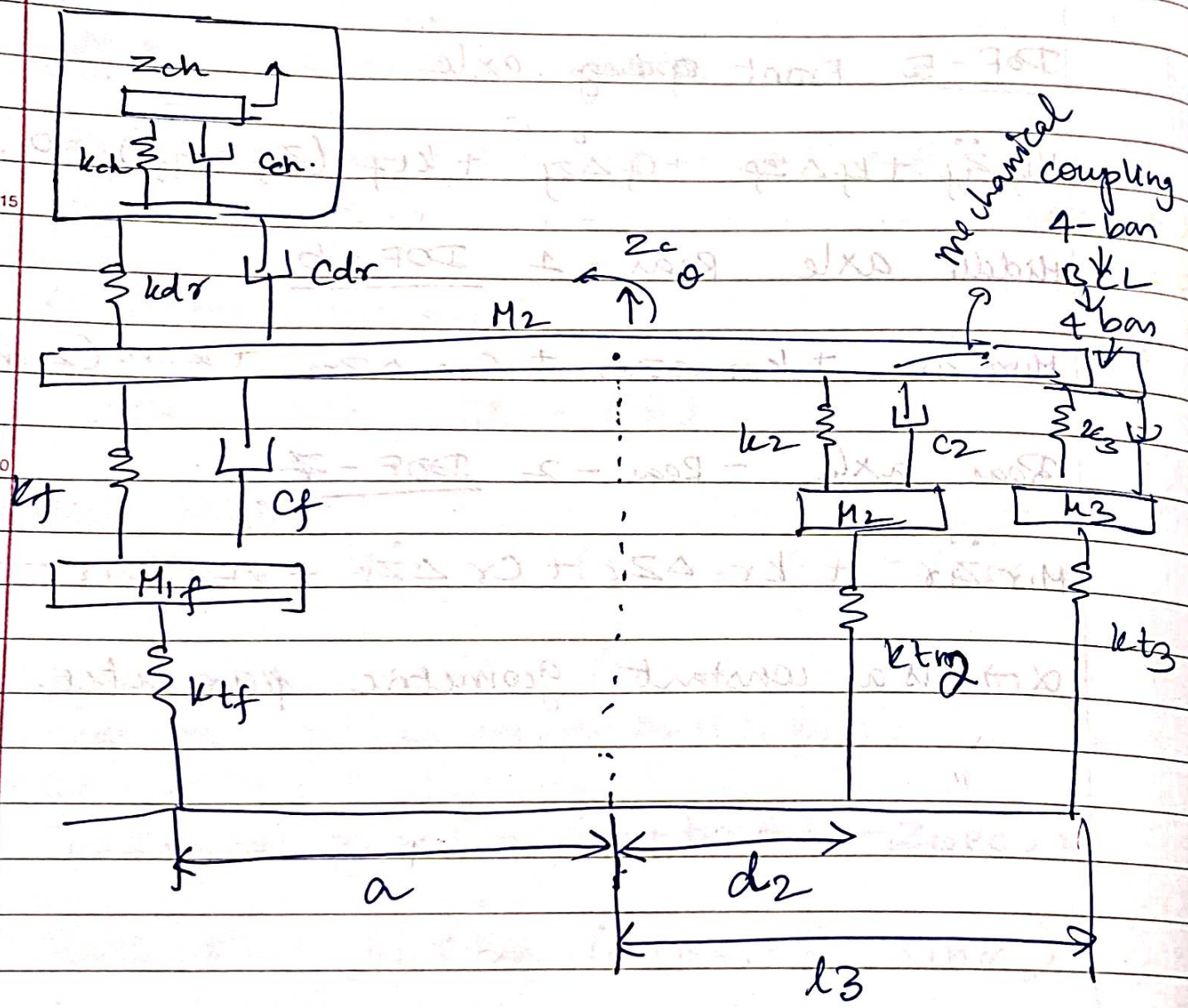
$$\alpha_f = \delta - \tan^{-1}\left(\frac{v_y + ar}{u}\right)$$

$$\alpha_m = -\tan^{-1}\left(\frac{v_y - l_2}{u}\right)$$

$$\alpha_r = -\tan^{-1}\left(\frac{v_y - l_3}{u}\right)$$

Tyre model

$$F_{yi} = \begin{cases} C_{\alpha i}\,\alpha_i & |\alpha_i| \leq \alpha_{coltic,i} \\ \mu F_{zi}\,\mathrm{sgn}(\alpha_i) & |\alpha_i| > \alpha_{coltic,i} \end{cases}$$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